



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

PURE MATHEMATICS
PAPER 1

6042/1
3 hours

NOVEMBER 2024 SESSION

Additional materials:
Answer paper
Graph paper
List of Formulae MF7
Scientific calculator [Non-Programmable]

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided on the answer booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 120.

If a numerical answer cannot be given exactly and the accuracy required is not specified in the question, then in the case of an angle it should be given correct to the nearest degree, and in other cases it should be given correct to 2 significant figures.

Answer **all** questions.

This question paper consists of 4 printed pages.

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[Turn over



- 1 Simplify $\frac{(x^2)^{\frac{1}{3}} \times \left(x^{-\frac{1}{2}}\right)^3}{x^{-\frac{1}{6}} \times x^{-\frac{2}{3}}}$. [4]
- 2 A geometrical progression has the first term a where $a \neq 0$, and a common ratio r where $0 < r < 1$. The sum to infinity of the progression is double the sum of the first eight terms.
- Given that the 9th term is 20, calculate the
- (a) value of r^4 . [3]
- (b) exact value of a . [2]
- 3 Y varies jointly as the square root of x and inversely as the cube of z . $Y = 16$ when $x = 9$ and $z = \frac{1}{2}$.
- (a) Express Y in terms of x and z . [3]
- (b) Find the value of z when $Y = 2.5$ and $x = 900$. [2]
- 4 Express $\frac{3-x+6x^2}{(1-x)(2+x)(1+2x)}$ in partial fractions. [5]
- 5 Given that the polynomial $h(x) = x^4 + 3x^3 + ax + 3$ is divisible by $x^2 - x + 1$ where $x \neq 0$.
- (a) Find the value of a . [4]
- (b) Hence factorise $h(x)$ completely. [2]
- 6 Solve the inequality $3|x - 2| > |2x - 1|$. [6]
- 7 Position vectors $\overrightarrow{OM}$ and $\overrightarrow{ON}$ are such that $\overrightarrow{OM} = 2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$ and $\overrightarrow{ON} = -\mathbf{i} - 2\mathbf{j} + 6\mathbf{k}$.
- Find
- (a) $\cos \widehat{MON}$, [3]
- (b) unit vector in the direction of $\overrightarrow{MN}$. [3]
- 8 The gradient of a curve at the point (x, y) is given by $\frac{dy}{dx} = 4xy^2$.
- Find the
- (a) general solution of the differential equation, [3]



- (b) particular solution of dy , given that $x = 2$ and $y = 4$. [3]
- 9 Prove by induction that $11^n - 4^n$ is a multiple of 7, for all $n \in \mathbb{Z}^+$. [6]
- 10 Given the equation $x^2 + y^2 - 3x^2y + 5y + 2x = 0$,
find
- (a) $\frac{dy}{dx}$, [3]
- (b) the equation of normal to the curve at the point (1;3). [4]
- 11 The function $f(x) = x^2 - 6x + 8$ is defined for $0 \leq x \leq 3$.
- (a) Find $f^{-1}(x)$. [4]
- (b) State the domain of $f^{-1}(x)$. [1]
- (c) Sketch the graphs of $f(x)$ and $f^{-1}(x)$ on the same axes. [3]
- 12 (a) Expand $\frac{2x+3}{\sqrt[3]{(1-\frac{x}{3})^2}}$ in ascending powers of x up to and including the term
in x^3 simplifying the coefficients. [6]
- (b) State the set of values of x for which the expansion is valid. [2]
- 13 (a) Show that the equation $\sqrt{x} + \sqrt{x+1} + \sqrt{x+2} - 5 = 0$ has a root
between 1 and 2. [4]
- (b) Using $x_0 = 1.5$ as the initial estimate, use the Newton – Raphson method
three times to estimate root of the equation to two decimal places. [7]
- 14 (a) Solve the equation $z^2 + 4z + 7 = 0$ giving the roots in the form
 $z_1 = a + \sqrt{bi}$ and $z_2 = -a - \sqrt{bi}$
Find
- (i) z_1 and z_2 [3]
- (ii) z_2 in the polar form. [4]
- (iii) $z_1 z_2$. [2]
- (b) Represent the complex numbers z_1 ; z_2 and $z_1 z_2$ on Argand Diagram. [3]



- 15 A circle has centre $C(-4; 5)$. The points $A(0; 8)$ and $D(-16; 8)$ lie on the circumference of the circle. Another point B has coordinates at $(0; 2)$.

- (a) Find the equation of the circle. [3]
- (b) Show that B lies on the circumference of the circle. [2]
- (c) Find the equation of the tangent to the circle at B . [3]
- (d) Calculate the coordinates of the midpoint of the chord BD . [2]
- (e) Hence or otherwise calculate the equation of the perpendicular bisector of BD . [3]

- 16 The table below shows values of x and y obtained from an experiment which satisfies the law $y = ab^{-x}$

x	1.0	1.5	2.0	2.5	3.0
y	4.0	5.7	8.0	11.3	16.0

- (a) Show graphically that these values satisfy the relationship $y = ab^{-x}$. [7]
- (b) Use the graph to find approximate values of a and b . [5]

